Debug the Cause

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Debug the Cause before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Debug the Cause trace and record the first step you can explain without help.

Use W2: Choose between two explanations for Debugging with revealing tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Debug the Cause explanation against the supplied facts before discussing it.

Use the worked case for Debugging with revealing tests. Proposed explanation: “The last wrong number locates the fault”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Debug a new ticket rule

A paper ticket rule must add 1 to the **input** count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5.

Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Debug the Cause target to a trace and a justified conclusion.

Starting at square 0, the target is square 3. Program: repeat twice {MOVE 1}; STOP. A teammate changes the target to 2 and calls the bug fixed. Repair the program and give two tests.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Debug the Cause case and show the deciding evidence.

A robot should go from square one to square three. Its program is MOVE ONE, STOP.

Find the cause and repair one thing.

Show your **trace**, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A sorting rule says blue goes left. A blue card goes right. The reader correctly reports blue, but the direction rule says blue goes right. Which part should you fix first?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the **expected** answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Debug the Cause adds to Helpful Classroom Sorter and keep the evidence.

When the sorter fails, log the first wrong stage and change only the responsible rule, **input**, or label.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

The requirement is to move exactly three squares forward from any valid starting square, then stop. Starting at square 0, repeat twice {MOVE 1}; STOP ends at 2. A teammate reduces the required distance to two squares.

Repair the program without changing the requirement; test starts 0 and 2.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Locate the first difference

> Locate the first difference. Apply the supplied Debug the Cause facts and keep your reasoning.

A procedure must add 2, double the updated value, then subtract 1. Starting at 3, an observed **trace** is 3, 5, 8, 7.

Write the **expected trace**, identify the first difference and explain why the final subtraction is not necessarily a separate fault.

> **My reasoning and evidence**

WORKBOOK · 13 OF 21

W2 Choose between two explanations

> Choose between two explanations. Apply the supplied Debug the Cause facts and keep your reasoning.

In W1 the faulty middle step may add 3, or add the original starting value. Both explain 5 becoming 8 when the start is 3. Choose start 1 or start 3 for a revealing new test.

Predict the final result under each explanation and under the intended procedure.

> **My reasoning and evidence**

WORKBOOK · 14 OF 21

W3 Reject a repair that hides the symptom

> Reject a repair that hides the symptom. Apply the supplied Debug the Cause facts and keep your reasoning.

A learner repairs the original points machine by reporting 12 for every **input**, because 12 is correct from start 4. Give a different starting **input** and its **expected output** under add 3, double, subtract 2.

Explain why the constant report is not a valid repair and what earlier **evidence** should be preserved.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Debug a new ticket rule

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Debug a new ticket rule. Apply the supplied Debug the Cause facts and keep your reasoning.

A paper ticket rule must add 1 to the **input** count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5.

Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Make a defensible completion claim

> Make a defensible completion claim. Apply the supplied Debug the Cause facts and keep your reasoning.

After repairing W4, all three chosen tests pass. Write a short report with the fault, the exact repair, the three **input**s and **expected output**s, and what those passing tests establish.

Explain why saying every possible error is now impossible would go beyond the **evidence**.

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Debug the Cause

> A fictional paper calculator should double the current value, but a copied instruction behaves strangely. Explain the labelled step before checking the changed condition.

A fictional paper calculator should double the current value, but a copied instruction behaves strangely.

After **input** 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double.

Both hypotheses fit this first **observation**. A new **input** may be chosen.

AI ACADEMY / AI-1/12

A revealing test separates explanations

	Input	Add 6 predicts	Double predicts
Original input	6	12	12
Recorded new input	2	8	4
Proposed input	0	6	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/12

Debug the Cause

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Observe
Input 6 produced 12; that fact fits two explanations.



2. Predict alternatives
For input 2, adding 6 predicts 8; doubling predicts 4.



3. Run revealing test
The fictional new record is input 2 -> output 8.



4. Locate cause
This record supports add-6 over doubling for this comparison.

CHANGE ONE THING
Use input 6 again instead of 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Propose **input** 0 and give both predicted **output**s without inventing an **observed result**.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a **trace** of how one **input** becomes an **output**; explain the rule or learned component.

When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper calculator should double the current value, but a copied instruction behaves strangely.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? This record supports add-6 over doubling for this comparison.

1. Observe

2. Run revealing test

3. Predict alternatives

4. Locate cause

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Locate cause This record supports add-6 over doubling for this comparison. Repeating **input** 6 would not separate the two hypotheses because both again predict 12.

Input 2 makes their **prediction**s different before the test is run. The second record supports investigating an added constant, but inspecting the instruction is still needed to confirm the implementation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use **input** 6 again instead of 2. Which conclusion is supported by

the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Both hypotheses predict 12; the test remains inconclusive.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both hypotheses predict 12; the test remains inconclusive. A revealing test requires different predicted **output**s, not merely another recorded run.

Repair a missing building block

Review Repeating Efficiently (ai-1/11). Goal: **Trace** a loop by testing its condition before every repetition. Short practice: **Trace** a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: **Trace** an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

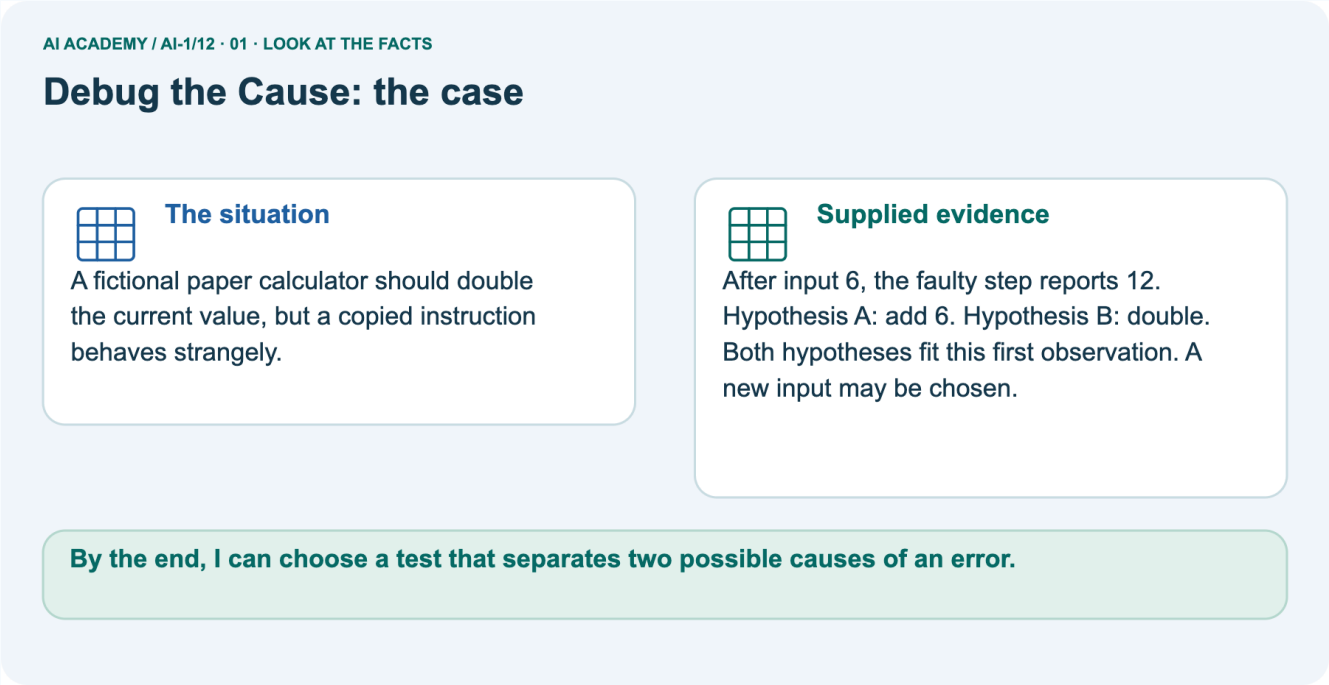
> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Debug the Cause picture from its labels, then record a relevant limit.

01. Debug the Cause: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

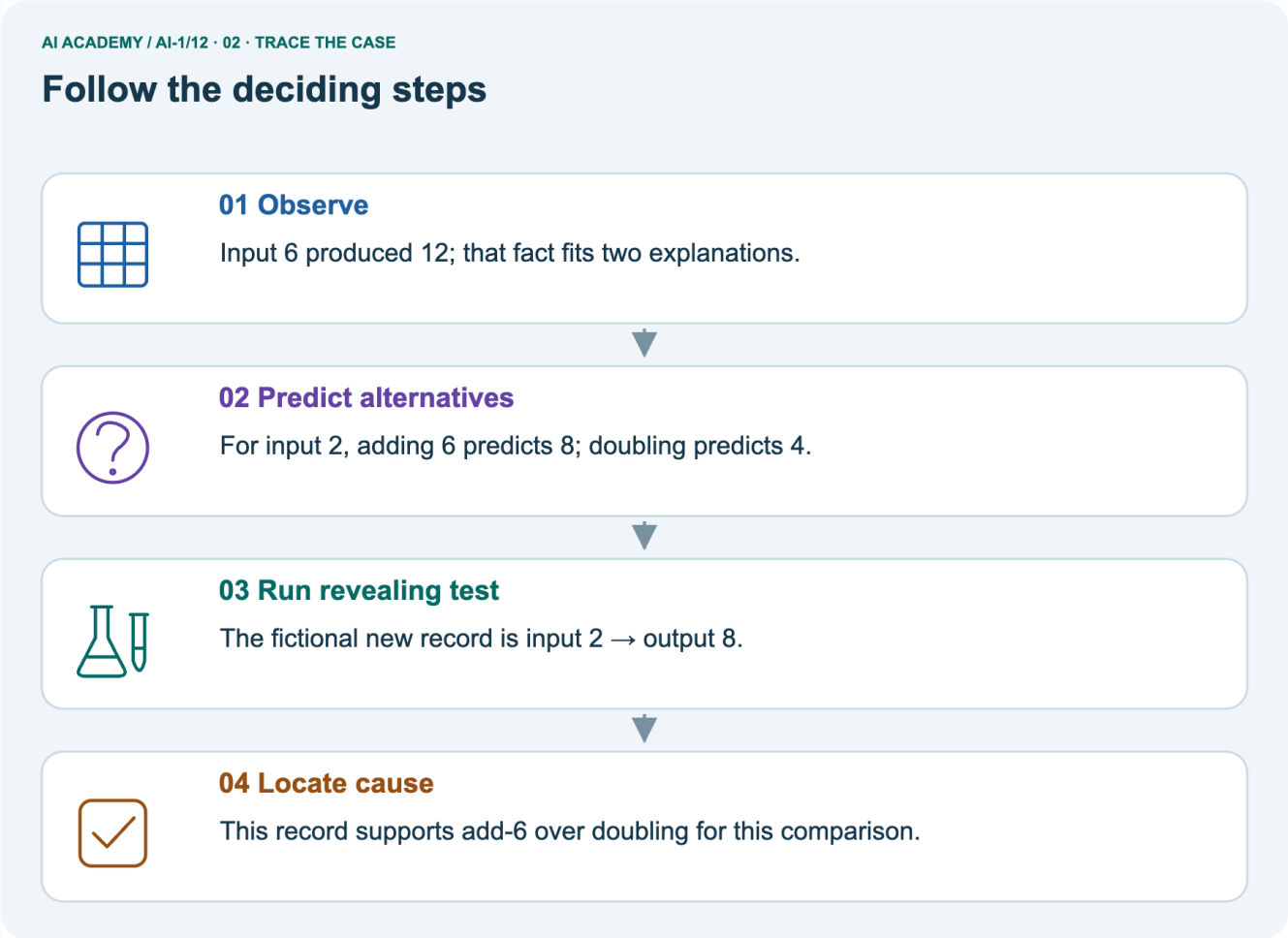
A fictional paper calculator should double the current value, but a copied instruction behaves strangely. Supplied facts: After **input** 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double.

Both hypotheses fit this first **observation**. A new **input** may be chosen.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



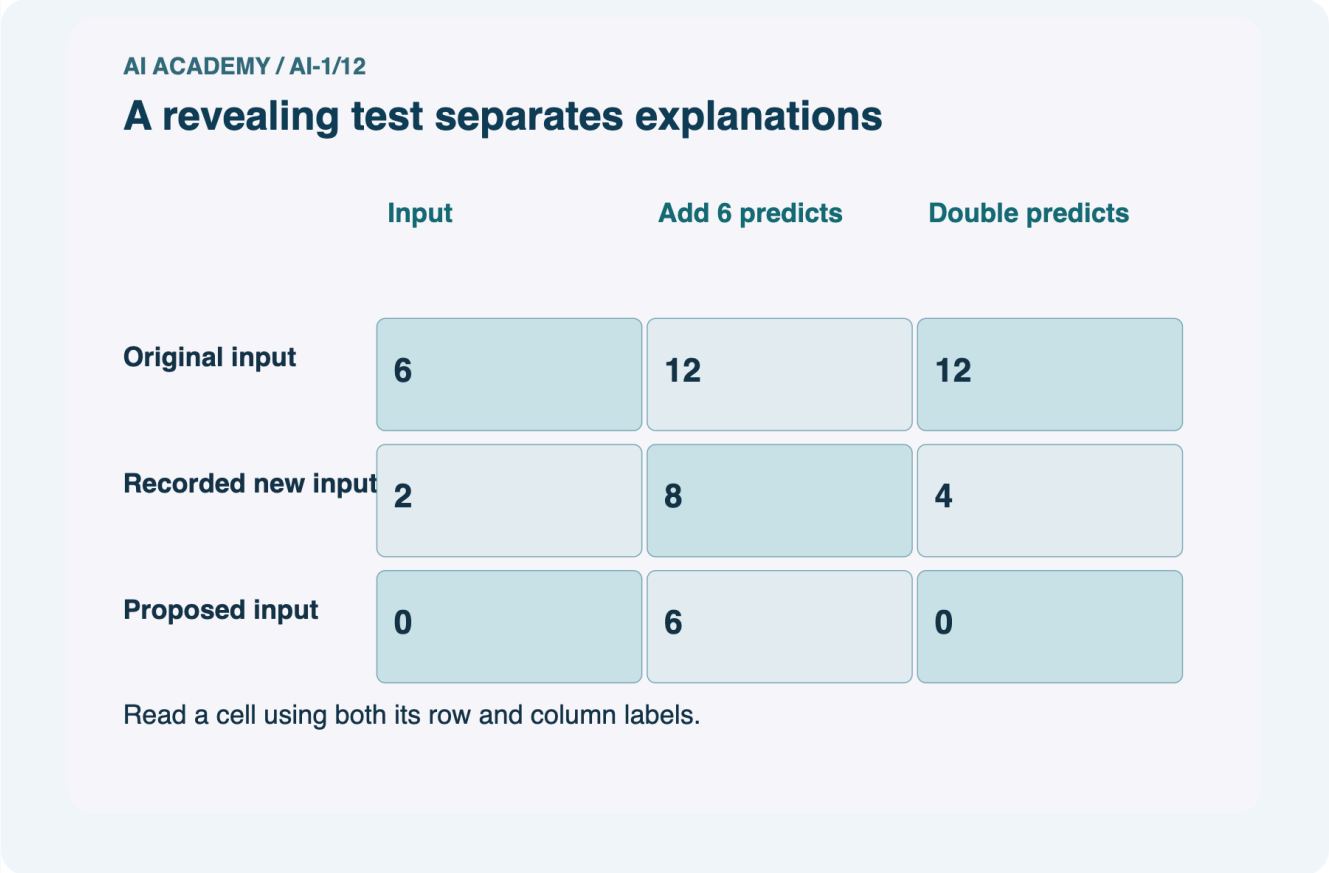
Follow the labelled picture with the matching explanation. Open the vector diagram

Observe: **Input** 6 produced 12; that fact fits two explanations. → Predict alternatives: For **input** 2, adding 6 predicts 8; doubling predicts 4. → Run revealing test: The fictional new record is **input** 2 → **output** 8. → Locate cause: This record supports add-6 over doubling for this comparison.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A revealing test separates explanations



Follow the labelled picture with the matching explanation. Open the vector diagram

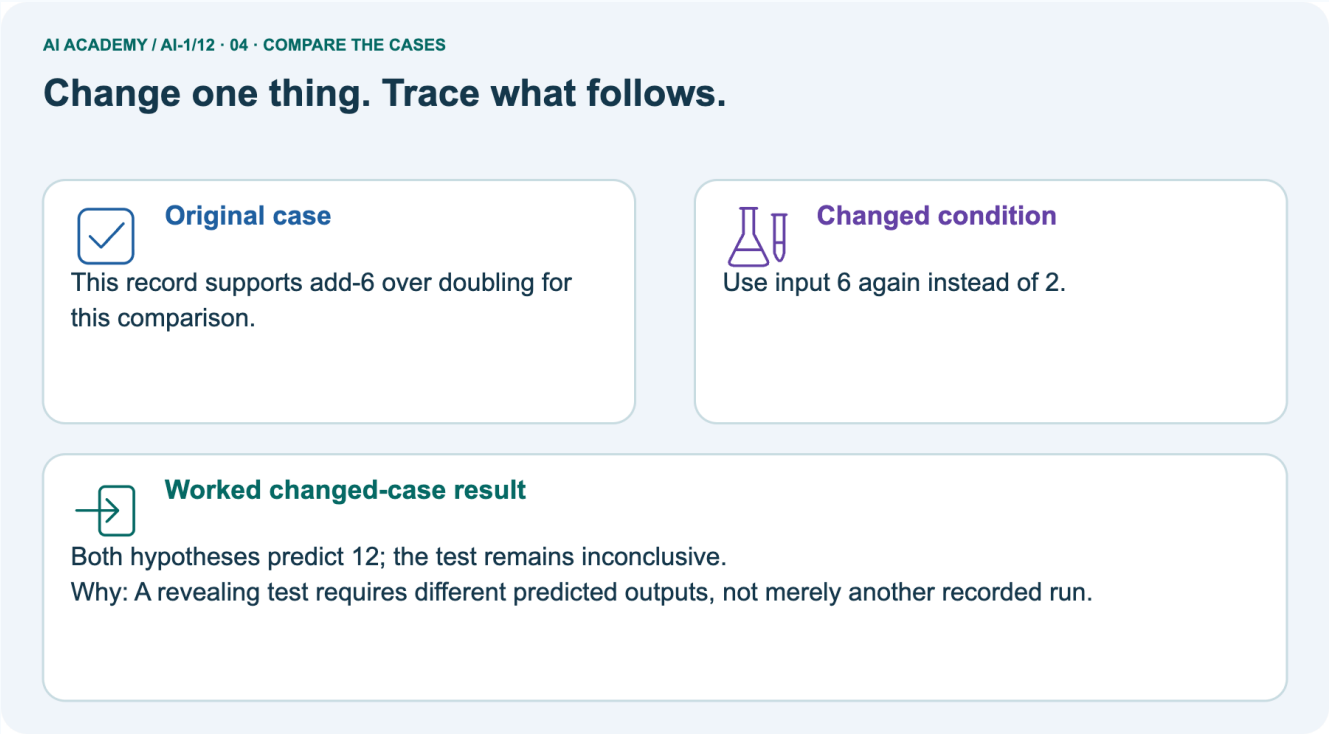
A revealing test separates explanations. Original **input**: **Input** = 6, Add 6 predicts = 12, Double predicts = 12. Recorded new **input**: **Input** = 2, Add 6 predicts = 8, Double predicts = 4.

Proposed **input**: **Input** = 0, Add 6 predicts = 6, Double predicts = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

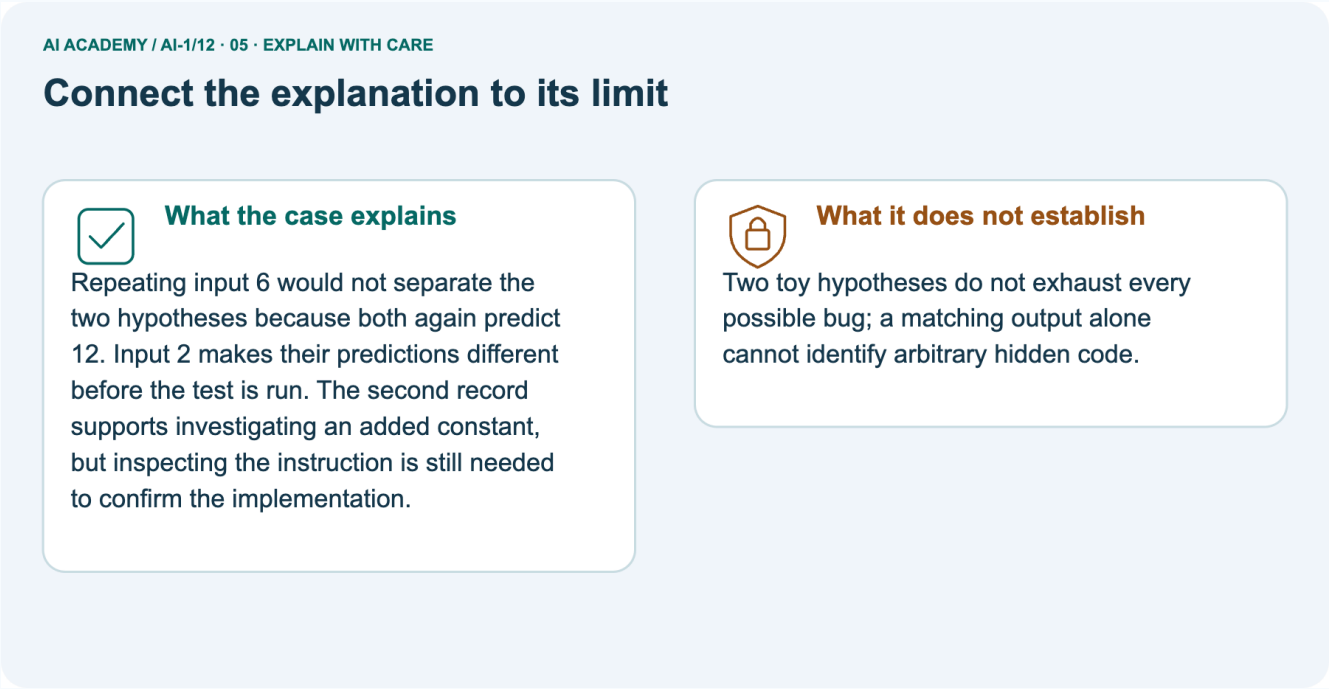
Original case: This record supports add-6 over doubling for this comparison. Changed condition: Use **input** 6 again instead of 2. **Expected result**: Both hypotheses predict 12; the test remains inconclusive.

Reason: A revealing test requires different predicted **output**s, not merely another recorded run.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Repeating **input** 6 would not separate the two hypotheses because both again predict 12. **Input** 2 makes their **prediction**s different before the test is run.

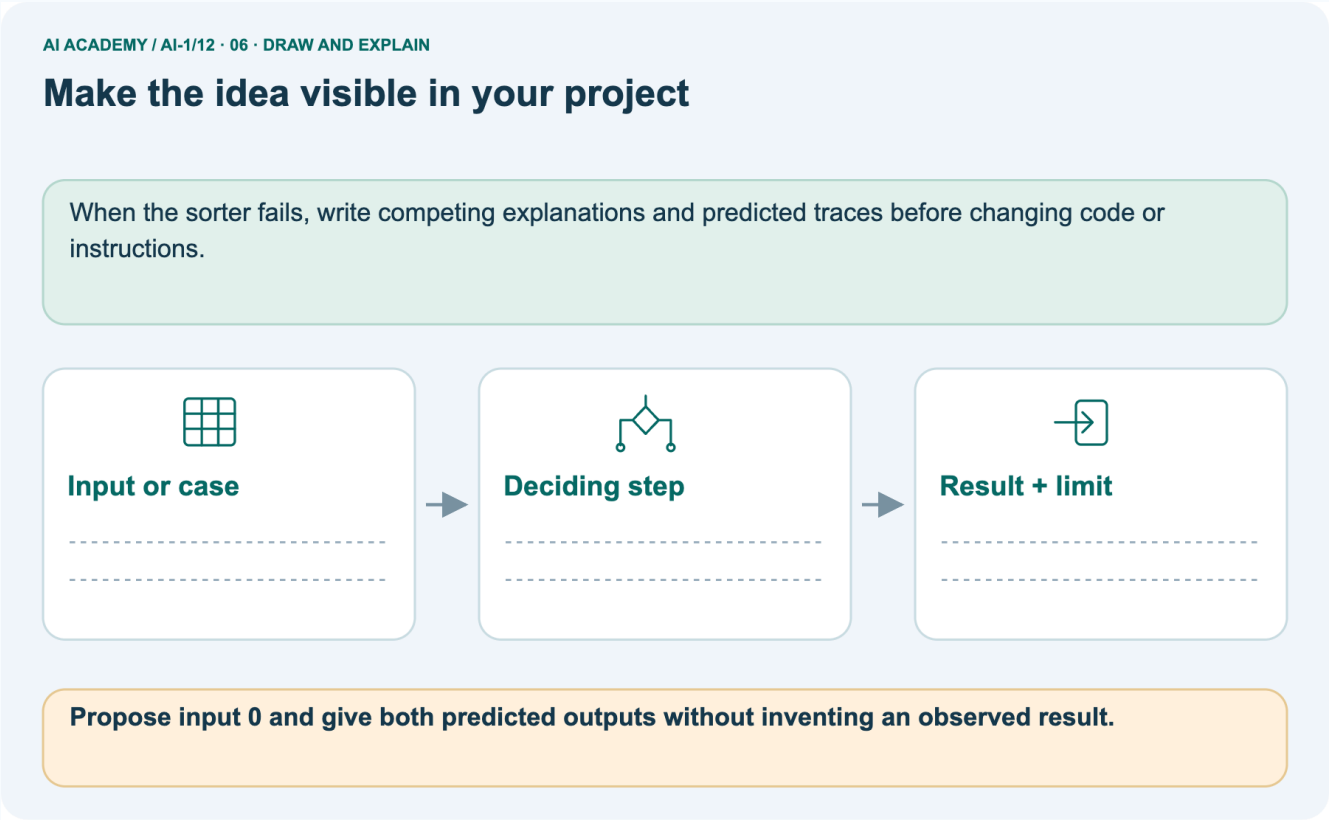
The second record supports investigating an added constant, but inspecting the instruction is still needed to confirm the implementation.

Limit: Two toy hypotheses do not exhaust every possible bug; a matching **output** alone cannot identify arbitrary hidden code.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions. Draw your **input**, deciding step, result and limit.

Supported question: Propose **input** 0 and give both predicted **output**s without inventing an **observed result**.

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.